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Roland S-1 “Tweak” Synthesizer

A pocket-sized poly that packs a punch

Brian Candler · Follow

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When searching for something to introduce my daughter to the world of synthesizers, I stumbled across the new Roland S-1. It’s the size and weight of a paperback book, runs off its own rechargeable battery, sounds amazing, and costs about a third of the other options I was looking at (such as the Korg Minilogue XD). What’s not to like?

Overview

The S-1 is a shrunk-down copy of the slightly larger SH-01A, which in turn is a digital recreation of the monophonic, analogue SH-101 which was sold from 1982 to 1986. Compared to its venerable ancestor it adds a host of features including 4-note polyphony, a sequencer and arpeggiator, and chorus/delay/reverb effects. Being a digital emulation, it never goes out of tune. What’s more, it can send and receive audio digitally via its USB-C port (which appears as an audio in/out interface on the host computer); the same port also doubles up for MIDI and charging.

It can store 64 “patterns” each containing both synth presets and sequencer data. The first 16 contain factory-set examples that you can overwrite, and the remaining 48 are blank.

Out of the box it comes with a USB-A to USB-C lead, and nothing else. A padded carry case is sold as an optional extra. There is no internal speaker, so to get started you’ll need to plug headphones into the 3.5mm output jack (unless you connect to a PC and take digital audio over USB). There’s not even a printed manual included — it’s a PDF download.

Keyboard

The S-1 has 25 small buttons serving as a two-octave keyboard, and which double up with other functions when used in conjunction with the Shift key. For example, Shift and Oct-/Oct+ (which are two of the black notes) can be used to move the keyboard up and down in octaves.

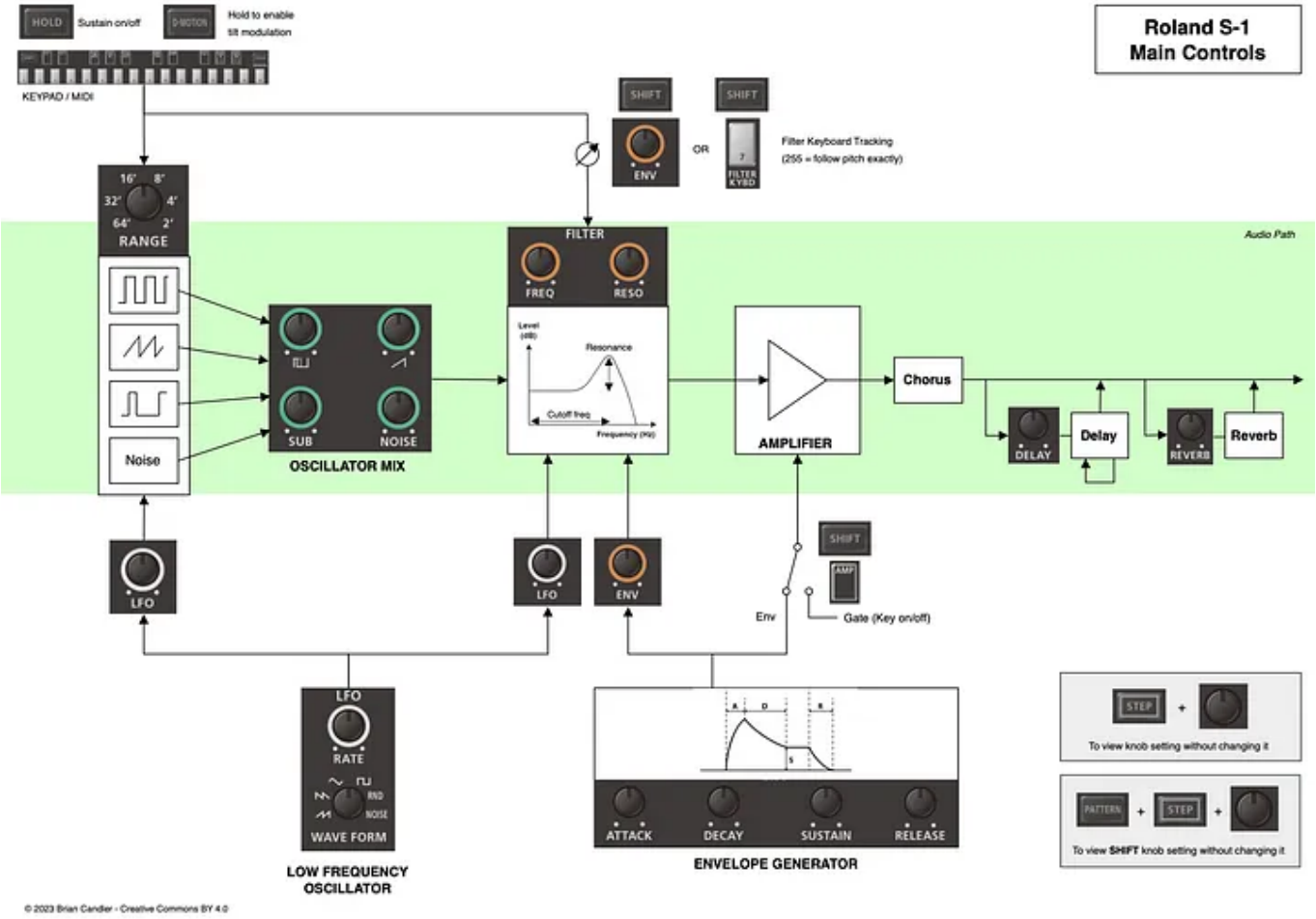
The keyboard is quite playable and certainly fine for setting up riffs in the sequencer; and of course you can always play it over MIDI. There’s a MIDI interface in the USB-C port, as well as MIDI in/out on 3.5mm jacks (a format

I’ve never seen before, but you can get adapters to convert this to regular MIDI ports)

There are no pitch bend or modulation wheels. Instead, there is a feature called “D-MOTION”: press a button, then tilt the whole device front-to-back or side-to-side. It makes the unit behave like a large joystick, and is lots of fun to use. The two directions can be separately routed to a range of different controls[1].

Audio path

Here is a simplified block diagram. It’s part of a 4-page “quick reference” that I created, which you can download from here.



Roland S-1 Main Controls — simplified block diagram

There are the traditional synthesizer components you’d expect:

- Three oscillators (square/pulse, triangle, and a square/pulse sub-oscillator which is 1 or 2 octaves below) plus a pink/white noise generator.
- A low-pass filter, with adjustable cutoff and resonance
- An ADSR envelope generator, which can control amplitude, filter cutoff, and/or pulse width modulation of the main square/pulse oscillator
- A low-frequency oscillator (modulation source) which can affect pitch, filter cutoff, and/or pulse width modulation. It includes a “random” shape for sample-and-hold-like effects.

These are followed by digital chorus, delay (echo) and reverb effects.

There’s no ring modulation, oscillator sync or FM synthesis, but you can run the LFO in high-speed mode which allows a range of effects to be achieved.

There are four polyphony modes: polyphonic, monophonic, unison (where all four voices play the same note for a rich layered sound), and chord (where you configure the intervals between the voices).

The filter is low-pass only and has a very peaky resonance, which can go into self-oscillation if you crank it all the way up — effectively giving you another sine-wave source. You can set up keyboard tracking so that the filter cutoff follows the input note — and incidentally, in polyphonic mode this demonstrates that there are 4 separate filters, rather than all four voices going into a shared filter, as some synths have.

However, this is the point where you realise the issues with the S-1’s small form factor.

Control surface

Due to the tiny size of the S-1, there are a limited number of control knobs which are directly available on the surface . They are quite fiddly, although still much more tactile and immediate than using a mouse with a software synth emulator.

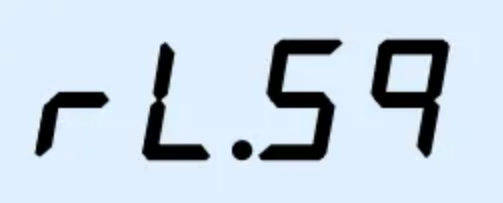
Some other controls are available[2] using Shift+knob combinations: for example, Shift + Filter Env knob (normally “filter envelope tracking”) controls filter keyboard tracking. These are not marked, so you have to learn where they are.

Other options are available using Shift + note keys, which are labelled. For example, Shift + AMP (one of the black notes) switches the amplifier section from envelope tracking to simple gate on/off control and back again.

But many other options are buried in menus, which you access using Shift + 15 (Menu), then use the Tempo/Value knob to select a menu, then hit Enter to go into that menu, adjust with Tempo/Value again, hit Exit to leave that setting and Exit again to leave the menus. Ugh. Chorus and LFO key sync are two examples which can only be controlled via menus.

And this is where the most unwelcome throwback to the 1980’s comes in: the 4-digit, 7-segment display, like an old alarm clock. It *tries* to display text here, and fails miserably. For example, what do you think this menu item is?

It’s actually trying to say “Mod.d” for “(LFO) Modulation Depth”. But that’s an easy one. How about this?



That’s “Reload Sequence” — restore the sequencer notes (but not the sound settings) from the saved pattern. How about these?

These are used when setting up chord mode. They are “v2.SW” (voice 2 switch on/off), and “v2.KS” (key shift, i.e. note offset) respectively.

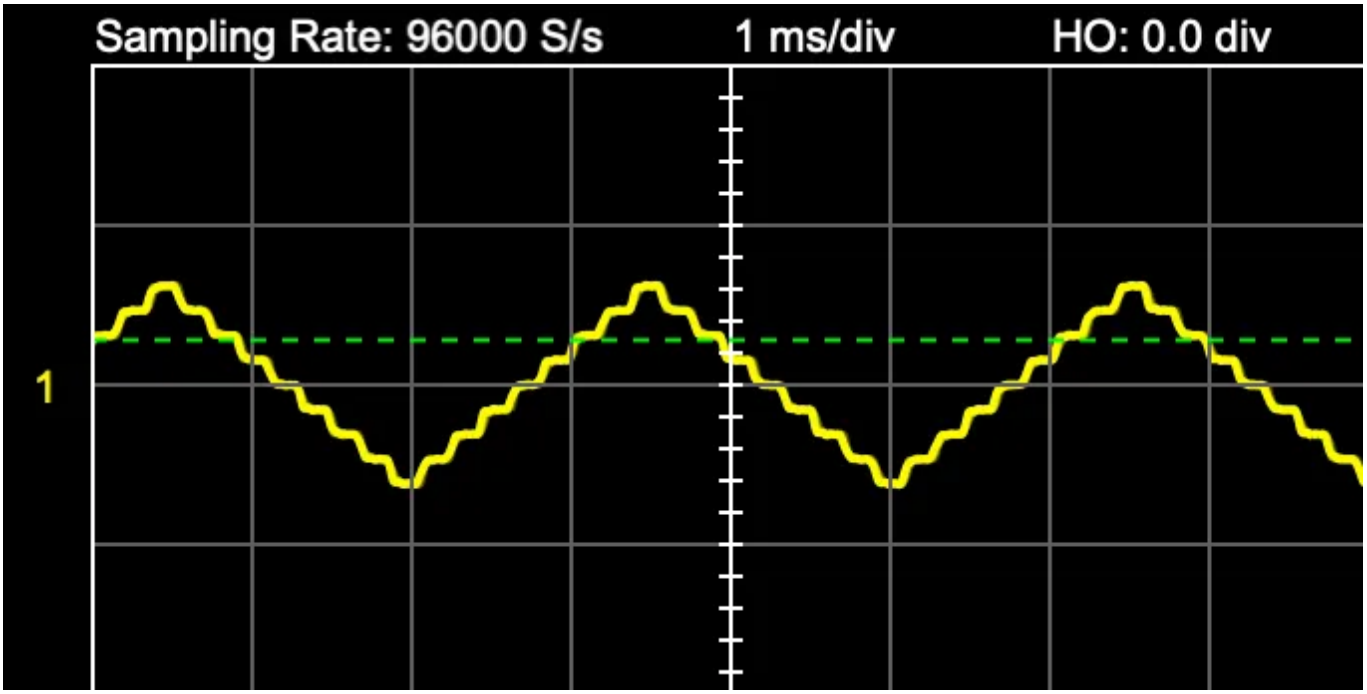
Roland could easily have fitted a dot matrix display in the space available. You can either see this as a ludicrous cost-saving measure to shave a few pence off the price; or you can see it as benefit, increasing its retro “nerd

appeal”. But it certainly doesn’t help if you’re trying to use the S-1 to introduce synthesizers to a newcomer. After a brief demo, my daughter gave a very succinct summary: *“it’s complicated”*.

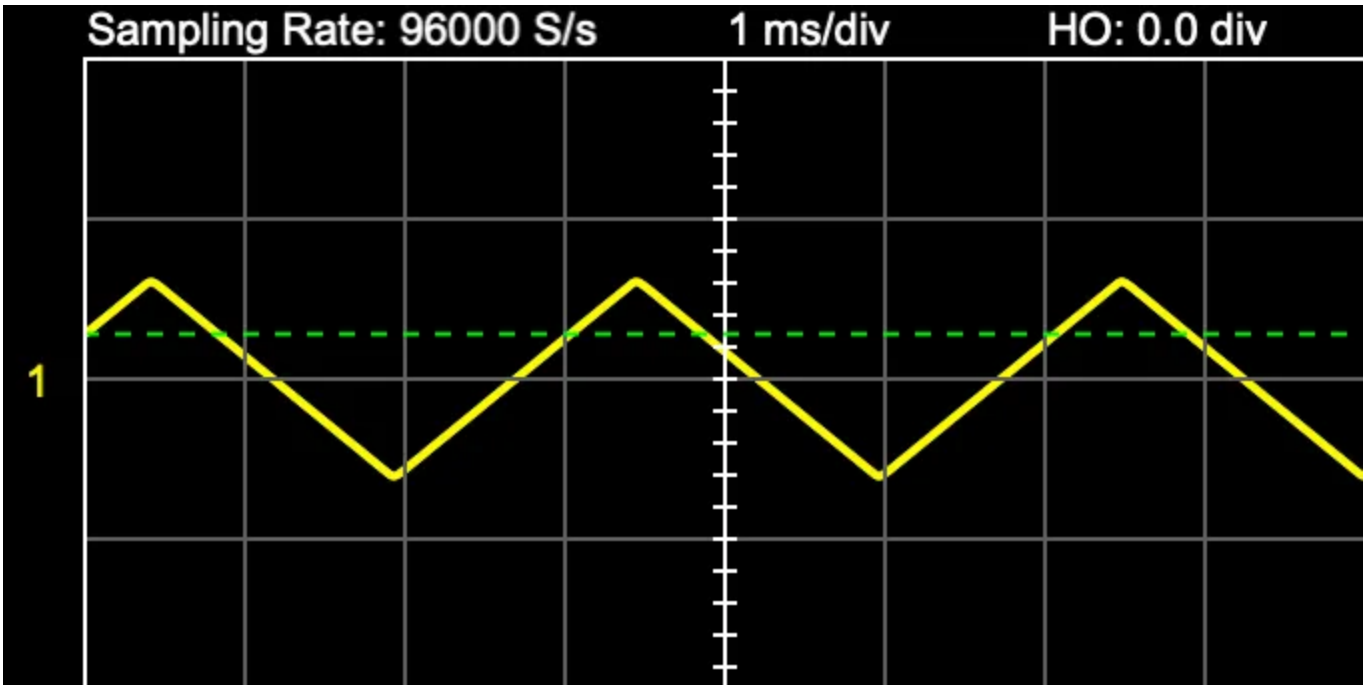
The manual covers everything, but you really need to work through it all and keep referring to the tables in the back. That’s why I ended up making the quick-reference card.

The lack of a dot-matrix display becomes a further drawback when you use some of the S-1's more advanced features. The square oscillator can be switched into an “Osc Draw” mode, which lets you plot your own waveform from scratch; and “Osc Chop” allows chopping an existing waveform to add extra harmonics. But these are very hard to use without a waveform display to see what you’re doing.

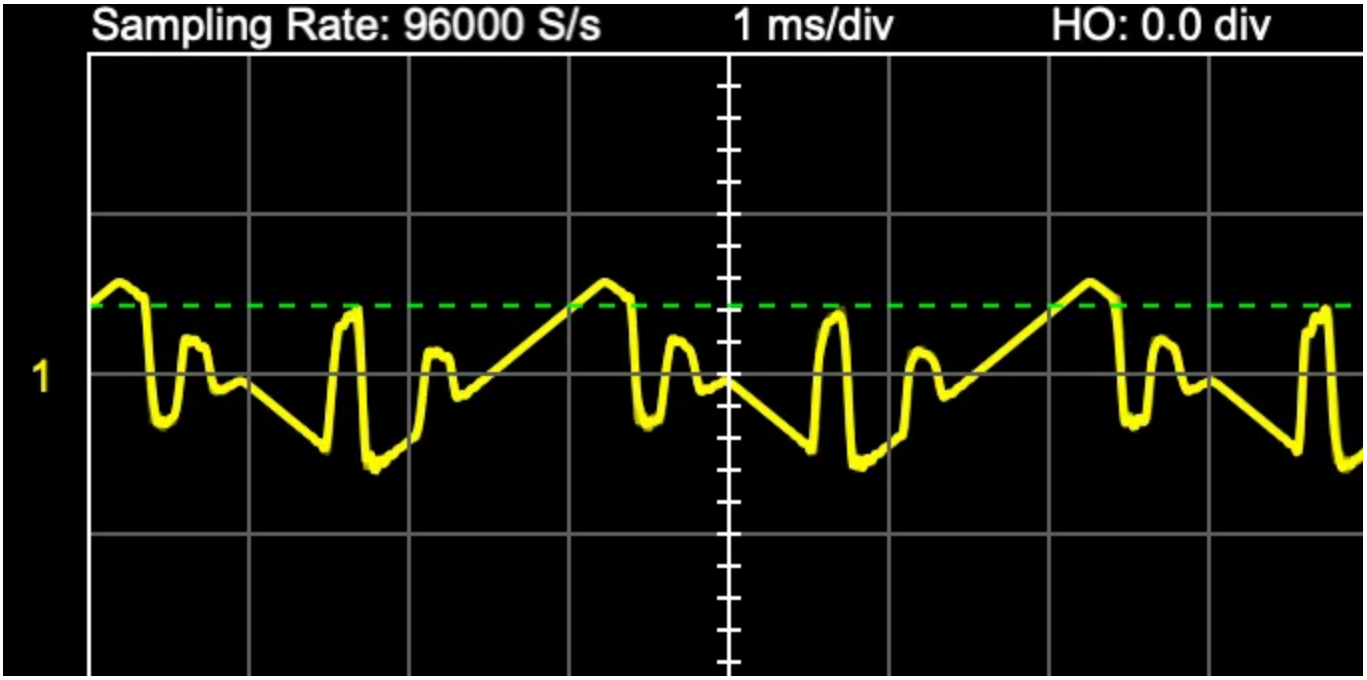
Fortunately, the USB audio capability provides a solution: you can connect to a PC and run a software oscilloscope, such as “Oscilloppoi” (99p from the Mac app store)



Oscilloppoi display of a triangle wave in Oscillator Draw “Step” mode



“Slope” mode smooths the edges



Adding some gritty “Chop” to the waveform

It it worth it?

The S-1 retails for around £180, and at this price it’s got to be one of the cheapest introductions to the world of “real” synthesizers you can get. It’s small enough to throw in a bag, and yet it’s a fully fledged instrument that wouldn’t be out of place in a professional studio. The MIDI, USB audio and Clock Sync connectivity means it can integrate into a digital audio workstation or any larger studio setup.

The Korg range of Monologue / Minilogue / Minilogue XD would, in my opinion, be much better for teaching and learning. They all have much better laid out control surfaces, and a built-in oscilloscope display. However, the Monologue is over 40% more expensive than the S-1, whilst only being a monophonic synth; and the other two are far more expensive, although you do get a proper 3 octave keyboard with these. They still have menus that you have to navigate for hidden settings. For example, only the Minilogue XD has a dedicated Portamento knob.

On the whole, the set of features you get from the S-1 in a compact and low-cost package is compelling.

If you want a more detailed look, I recommend this video by “Jorb”, a genuine synthesizer nerd. Just be warned that he covers it at break-neck speed!

. . .

[1]Roland refers to the two directions as “Pitch” and “Roll”, like movements on an aeroplane. That’s technically correct but a bit confusing, as “Pitch” means the front-to back movement, not musical pitch — although you can route this movement to pitch bend, should you wish.

[2]The range of shift controls has been expanded in firmware version 1.02. The unit I bought came with version 1.01, so I upgraded it over the USB port. This is a fairly straightforward operation.

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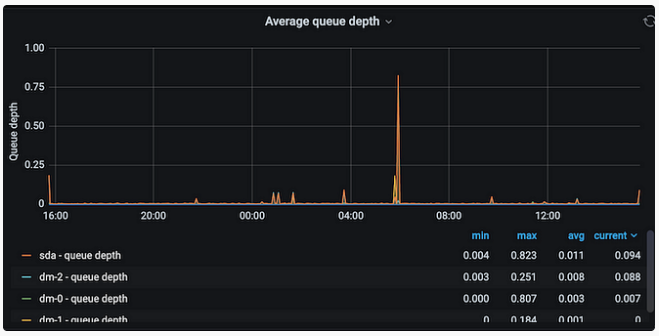

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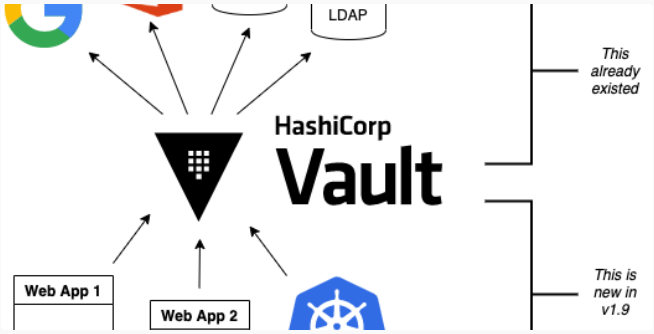

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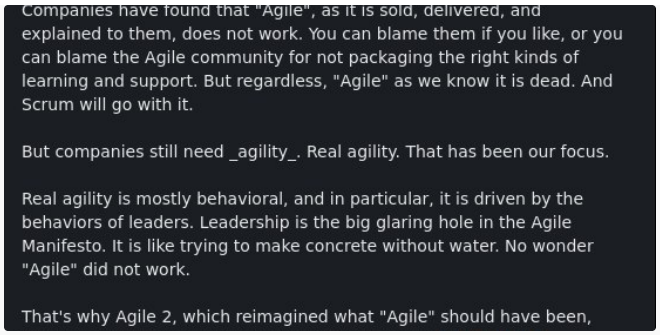
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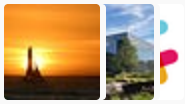

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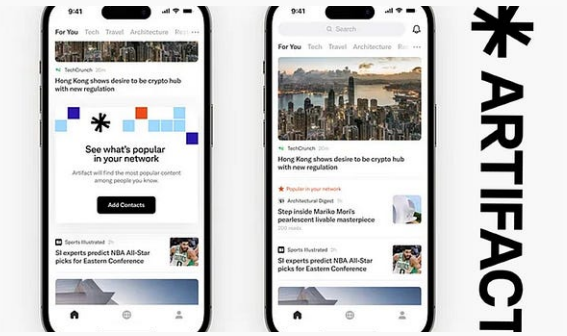

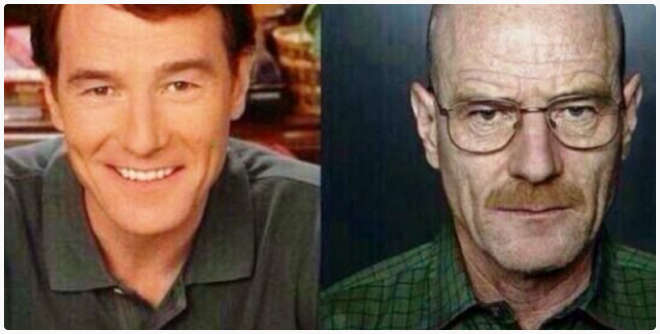

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